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BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

(Autonomous Institution Under VTU)

Yelahanka, Bengaluru -560119

INTERNSHIP REPORT

Duration: 9th April 2025 – 9th June 2025

Company Profile

Trinity Technologies Ltd

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GUIDE : MS.SPURTHY S N

About the Company

- Trinity Technologies Ltd is an AI and digital solutions company .
- Specializes in computer vision, IoT, edge computing, and cybersecurity innovations.
- Works on developing intelligent systems that improve safety, security, and efficiency in industries.



The logo for Trinity Technologies is positioned in the background of the top header. It features a stylized globe icon composed of various colored segments (blue, purple, green) and the company name 'trinity' in a lowercase, sans-serif font, with 'TECHNOLOGIES' in a smaller, uppercase font below it.

INTERNSHIP OBJECTIVES

- To gain practical experience in AI, computer vision, and cybersecurity simulation.
- To design and prototype intelligent systems for real-world industrial and educational applications.
- To enhance technical, analytical, and problem-solving skills in emerging technologies.

Project 1

Phishing Attack Simulation

- Designed a controlled phishing simulation to demonstrate how phishing attacks work.
- Aimed at raising awareness about credential theft risks and social engineering.
- Built the simulation using HTML, CSS, JavaScript, PHP, and MySQL.
- Highlighted how user data can be harvested if proper caution is not exercised.

Project 2

Computer Vision Tool for PPE Detection

- Developed an AI-based system to check if workers are wearing safety gear.
- Used YOLO (You Only Look Once) model fine-tuned for PPE detection.
- Integrated real-time video processing using OpenCV.
- Focused on edge deployment for fast, local detection and alerts.

Tools & Technologies Used

- Phishing Simulation: HTML, CSS, JavaScript, Python, MySQL & ngrok.
- PPE Detection: Python, OpenCV, PyTorch/TensorFlow, YOLOv5, Labellmg.
- Git/GitHub for version control.
- PowerPoint, Word for documentation and presentation.

Weekly Progress Summary

- Weeks 1-2: Orientation, requirement gathering, architecture planning.
- Weeks 3-4: Dataset preparation, model training (PPE project); Phishing site prototype.
- Weeks 5-6: Integration, testing, tuning of both projects.
- Weeks 7-8: Final report writing, PPT preparation, and project presentation.

Outcomes of Internship

- Strengthened practical skills in AI, computer vision, and cybersecurity.
- Gained experience in real-time system development and deployment concepts.
- Improved understanding of phishing risks and preventive awareness.
- Enhanced teamwork, report writing, and presentation abilities.

Conclusion

- The internship offered an excellent platform to apply theoretical concepts in practical projects.
- The experience helped build a strong foundation in intelligent system development and cybersecurity awareness.
- Prepared me for future roles in AI, computer vision, and ethical technology development.

THANK YOU

Any Questions ?